# GATE (ALL BRANCHES)

Engineering Mathematics

## Probability and Statistics





Lecture No. 01

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01 Fundamentals of Probability

02 Problem Solving Strategies

# Syllabus

✓ Module 01

# GATE – 13-15 marks

↑ for only CS

✓ Prob | statistics
✓ Linear Algebra
✓ Single variable Calculus

CS/ME/EE/EC
CE/ME/EI/IN
IT/

# IES/XE/

13-15 marks

Module II

→ Multivariable Calculus
→ vector Calculus
→ Differential equ$^{n}$ + Partial D.E
→ Fourier Series
→ complex Analysis
→ Numerical methods
→ Transfoon Theory

ME
CE
EE
EC
EI
IN

# Probability and statistics:

A) Basic Prob
B) classification of events
C) Bayes theorem + conditional Prob
D) Random variable
E) Prob. distribution
(F) correlation / regression

- Binomial
- bernoulli
- Poisson
- Exponential
- Normal
- Standard Normal
- Uniform Discrete
- Continuous

3-5 Marks

✓

Rahul Sir PW

Telegram

# Probability : Study of Uncertainity (Results Are uncertain)

# Experiment (Random)

# Tossing A Fair coin
OR
Tossing A Unbiased coin

# Sun Rise — East (Certain events)

# 1 Week = 7 days [certain events]

Fair coin ⟹ Mass Distribution SAME = Unbiased coin = prob. (part)

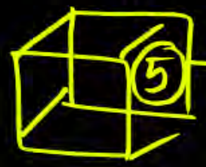
(TH)

Unfair coin ⟹ mass Distribution Different = biased coin

(T | H) due to gravity weight — (H)

# Tossing A Fair coin

# Throwing A Fair Die (Unbiased die) [Mass distribution equal]




→ weight (more) = Biased Die

A) Prob → Random experiment

# Random experiment → Results Are obtained Two or more ways

Tossing A fair coin

# Sample space or Sample Pts or SET

$S = \{H, T\}$

# Events → Head occurs / Tail occurs ] subSET

# Throwing A fair Die

$S = \{1, 2, 3, 4, 5, 6\}$

Events → (subSET) of Sample Space

SET { 1, 2, 3, 4, 5, 6

Prob. definition : $P(E) = \frac{\text{No. of Favourable outcomes}}{\text{Total No. of Possible outcomes}}$

$P[E] = \frac{\text{No. of desired outcomes}}{\text{Total No. of Possible outcomes}} = \frac{n(E) \leftarrow \text{Events}}{n(S) \leftarrow \text{Total}}$

Tossing A Fair coin

$S = \{H, T\}$

$P(\text{HEAD occurs}) = \frac{n(H)}{n(S)}$ $\quad P(\text{Tail}) = \frac{n(T)}{n(S)}$

$P(H) = \frac{1}{2}$ $\quad P(T) = \frac{1}{2}$

$P(\text{HEAD}) = 50\%$ $\quad P(\text{Tail}) = 50\%$

# Throwing A Die $S = \{1, 2, 3, 4, 5, 6\}$

$$P(1 \text{ occur}) = \frac{n(1)}{n(S)} = \frac{1}{6}$$

$$P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

# $P(H) = \frac{1}{2}$ $P(T) = \frac{1}{2}$

Tossing 5 Times

H H H H H

# 6 times Die – Throws

3 4 5 2 1 3

$P(6) = \frac{1}{6}$

$P(6) = 0$

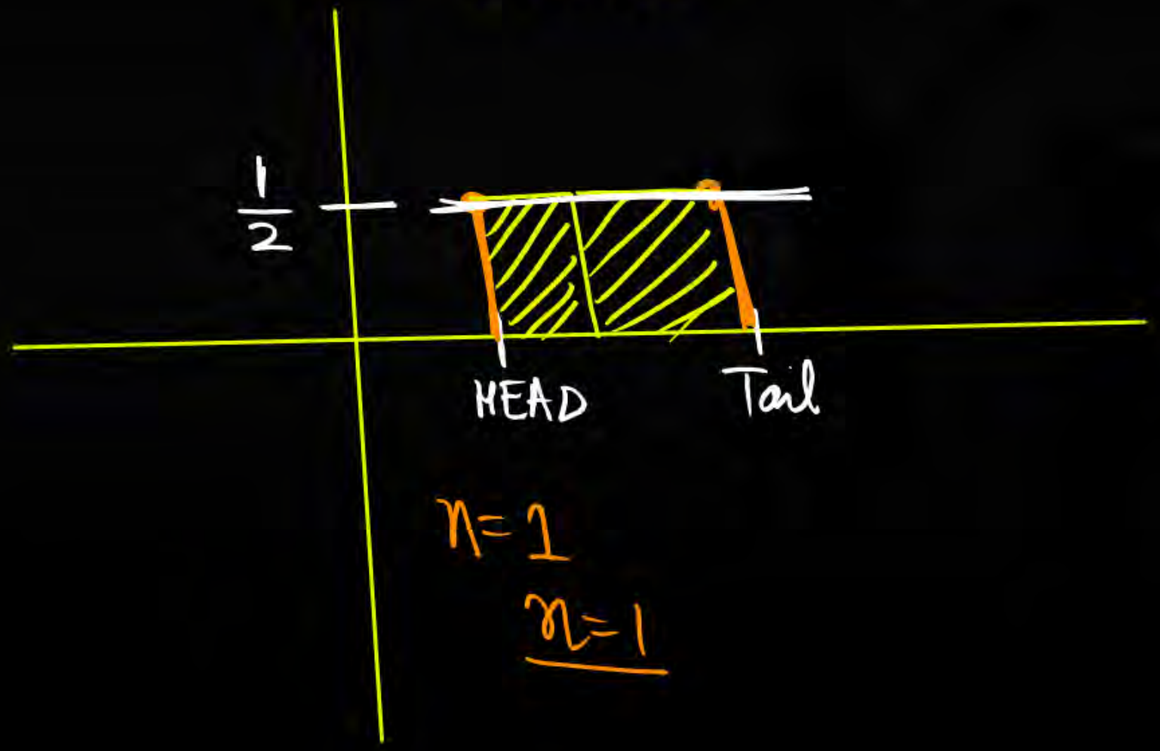
$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

# Tossing A Single coin

$S = \{H, T\}$ $P(H) = \frac{1}{2}$

A = HEAD occurs $P(T) = \frac{1}{2}$

B = Tail occurs

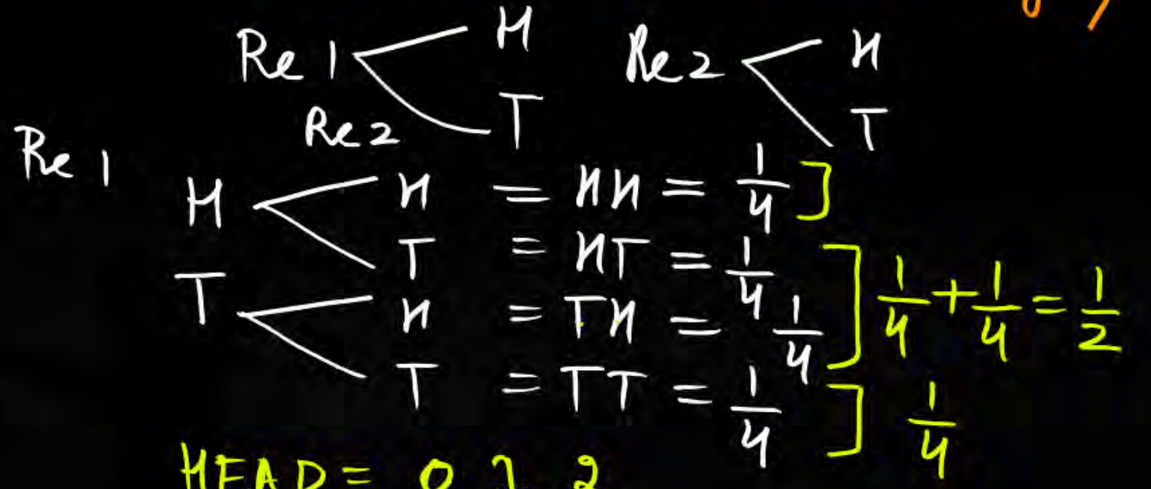



$n = 1$

$n = 1$

for $n = 2$

Re 1, Re 2 (simultaneously)




HEAD = 0, 1, 2

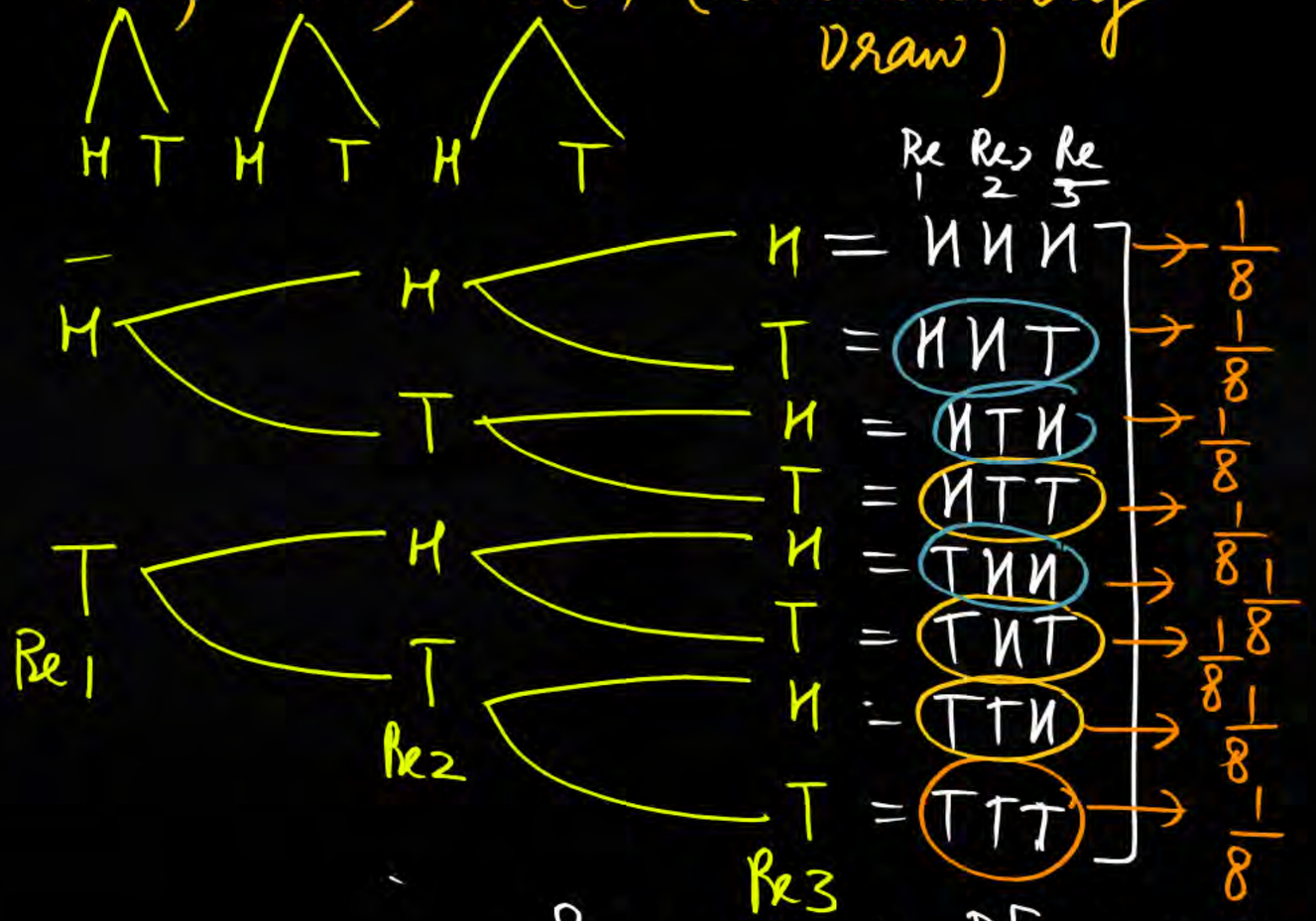
Tail = 0, 1, 2




$n = 2$

# $n=3$

Re 1, Re 2, Re (5) (simultaneously Draw)

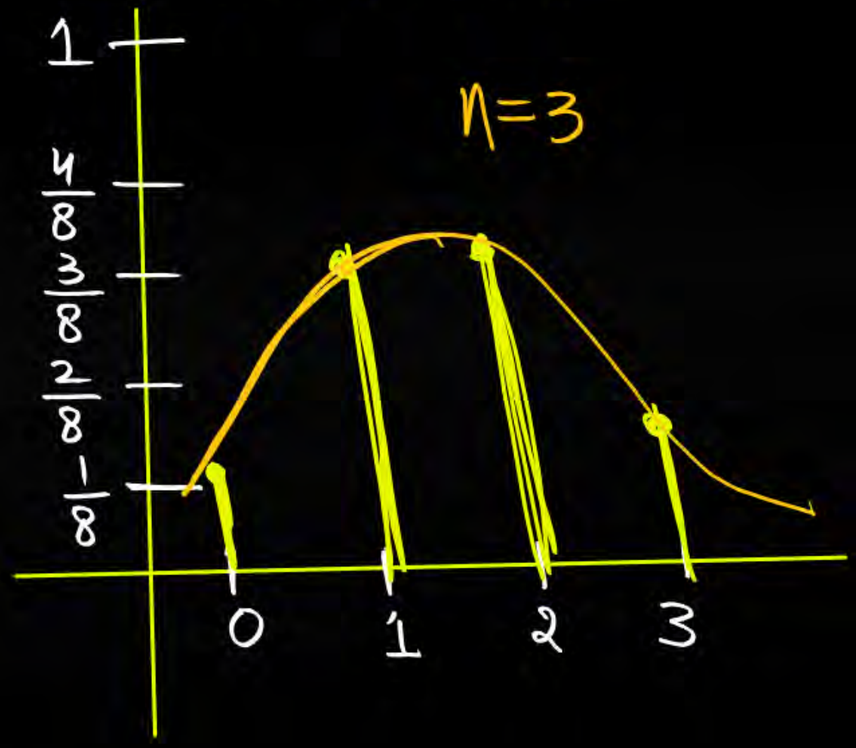



8 SAMPLE PTS

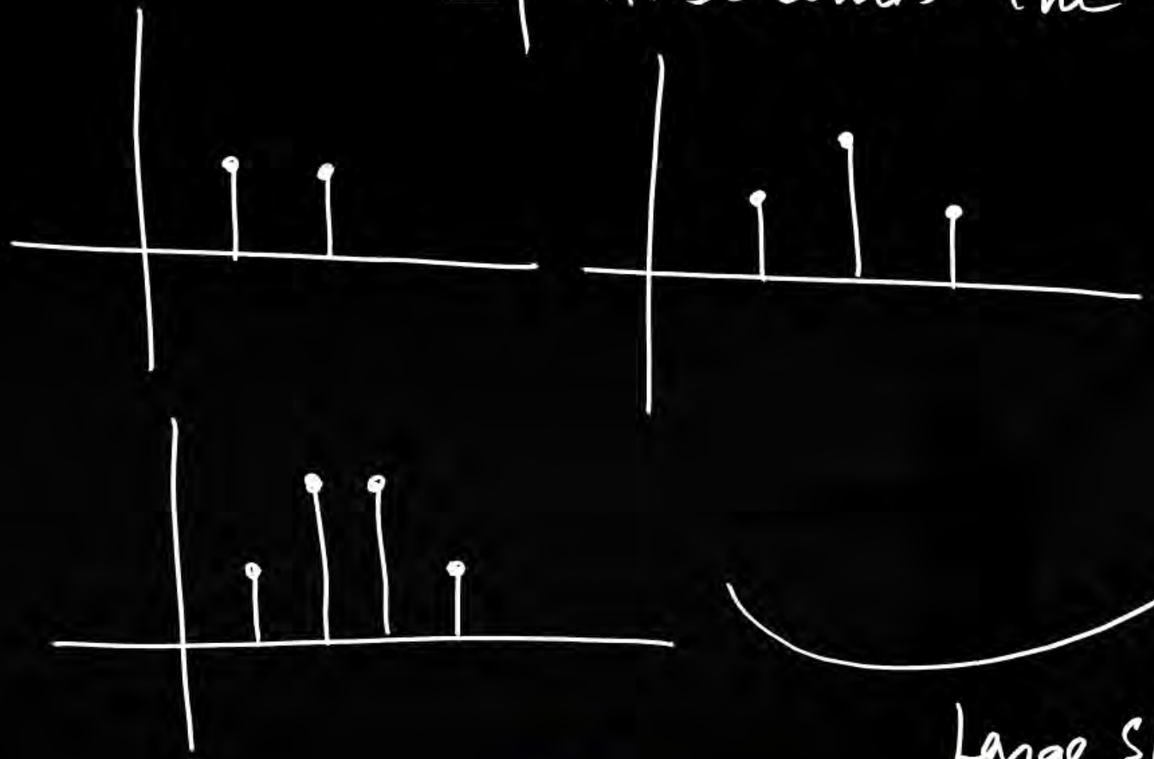
$P[0\text{HEAD}] = \frac{1}{8}$ $P(1\text{HEAD}] = \frac{3}{8}$

$P(2\text{HEAD}] = \frac{3}{8}$ $P(3\text{HEAD}) = \frac{1}{8}$

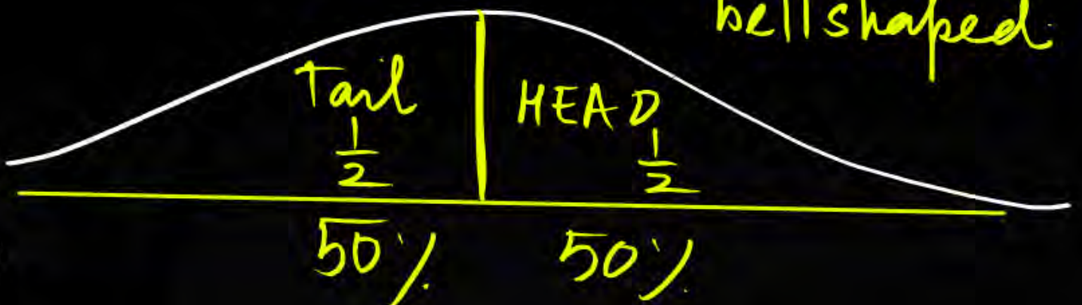
{ 0H, 1H, 2H, 3H
[ 0T, 1T, 2T, 3T

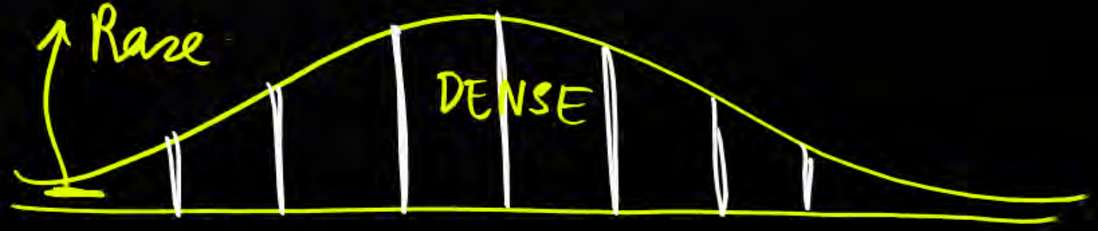


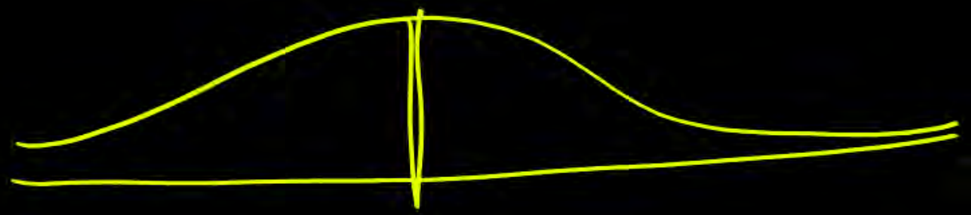
If $n > 30$ comes the curve convert to the bell shaped



Large SAMPle



If $n$ is large. Then
Curve is bell shaped.
[central Limit Theorem]

# Throwing A Two Die

Die A, Die B

If Events Are WORK
(simultaneously)
(working Together)

Working Together
= Die A X Die B
= 6 X 6
= 36 options

Three Die = Die A X Die B X Die C
= 6 X 6 X 6 = $6^3$ = 216 options

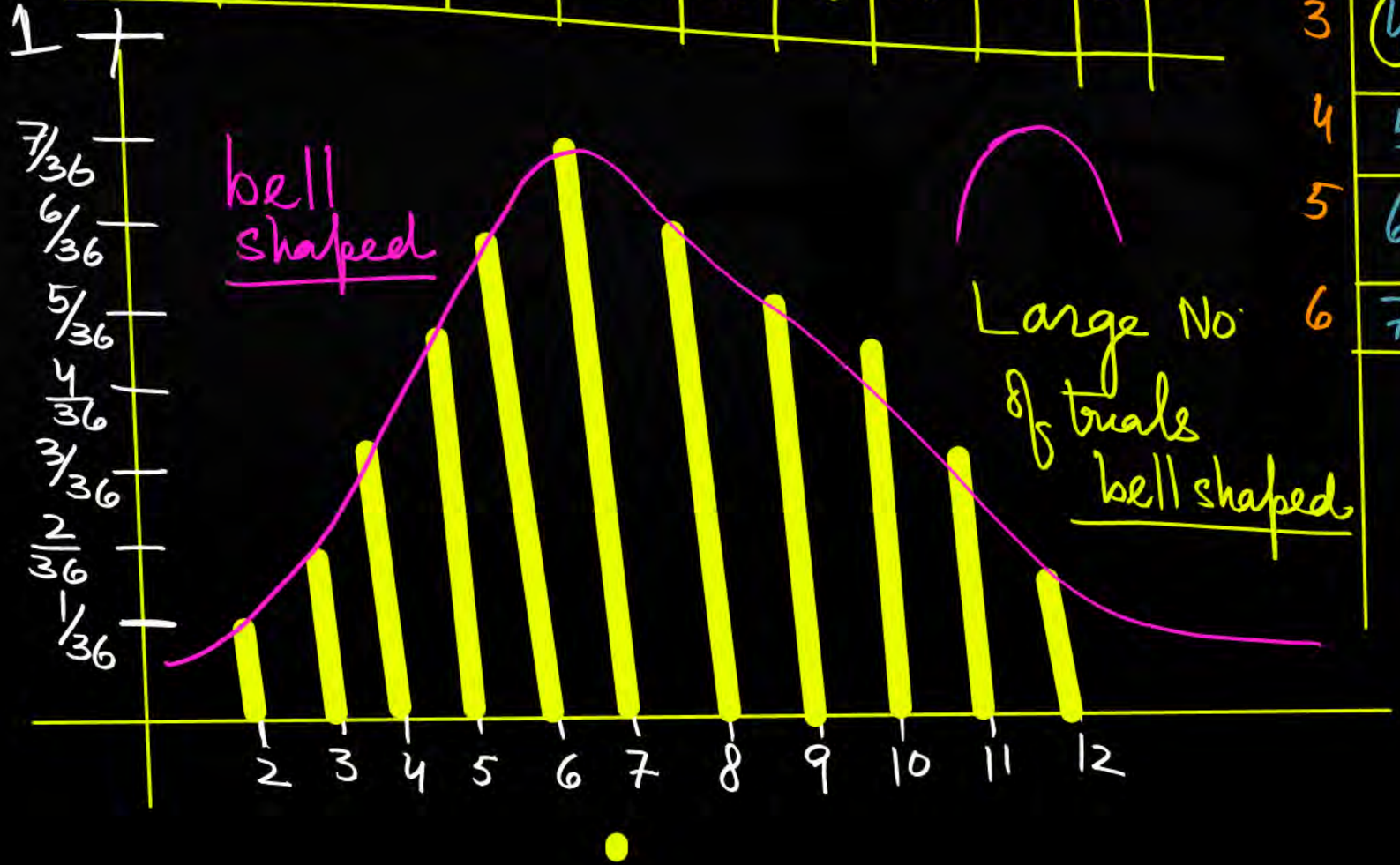
| Die A \ Die B | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | (1,1) | (1,2) | (1,3) | (1,4) | (1,5) | (1,6) |
| 2 | (2,1) | (2,2) | (2,3) | (2,4) | (2,5) | (2,6) |
| 3 | (3,1) | (3,2) | (3,3) | (3,4) | (3,5) | (3,6) |
| 4 | (4,1) | (4,2) | (4,3) | (4,4) | (4,5) | (4,6) |
| 5 | (5,1) | (5,2) | (5,3) | (5,4) | (5,5) | (5,6) |
| 6 | (6,1) | (6,2) | (6,3) | (6,4) | (6,5) | (6,6) |

$P(\text{SAME NO}) = \frac{6}{36} = \frac{1}{6} \Rightarrow$

# P(Die A + Die B) = graph = sum

| 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|
| $\frac{1}{36}$ | $\frac{2}{36}$ | $\frac{3}{36}$ | $\frac{4}{36}$ | $\frac{5}{36}$ | $\frac{6}{36}$ | $\frac{5}{36}$ | $\frac{4}{36}$ | $\frac{3}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ |

| A + B | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | (2) | (3) | (4) | 5 | 6 | 7 |
| 2 | (3) | (4) | 5 | 6 | 7 | 8 |
| 3 | (4) | 5 | 6 | 7 | 8 | 9 |
| 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 6 | 7 | 8 | 9 | 10 | 11 | 12 |




Large No of trials bell shaped

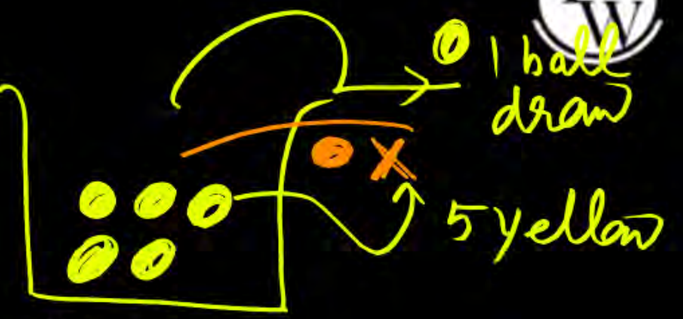
Total outcomes = 36

$P(2) = \frac{1}{36}$

$P(3) = \frac{2}{36}$

$P(4) = \frac{3}{36}$

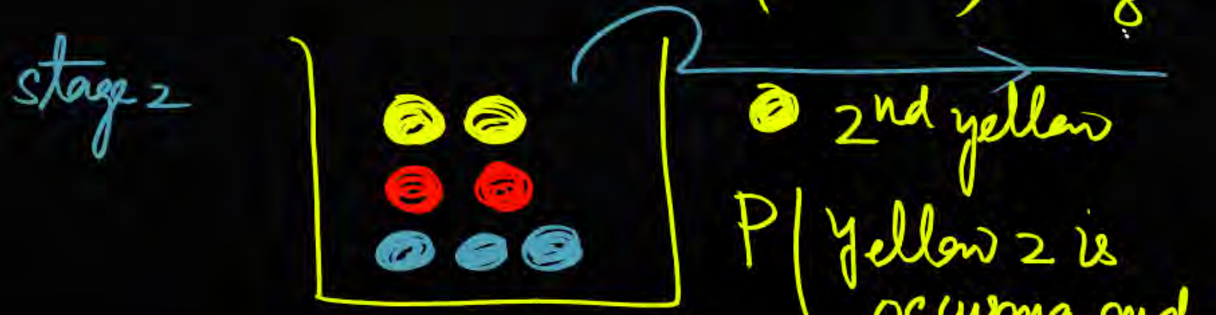
# Counting strategies : (Without replacement)
(Default)




A) What is The Prob.
P(2 yellow balls Are drawn at random one at a time)




$$P(\text{Yellow 1}) = \frac{3}{8}$$




$$P(\text{Yellow 2 is occuring and Yellow 1 is occured}) = P(\text{Yellow 2} \mid \text{Yellow 1}) = \frac{1}{7}$$

↑ occuring

↓ occured

(SAMPle space Are Decreased)

A) Chances/prob. Are change
B) Dependent events

$$P(\text{2 yellow ball}) = P(\text{yellow1}) \times P\left(\frac{\text{yellow2}}{\text{yellow1}}\right)$$

$$= \frac{3}{8} \times \frac{2}{7} = (\text{Working Together})$$

$$= \frac{6}{8 \times 7} \quad \underline{\text{Ans}}$$

Without Replacement
- → Dependent
- → Chance change
- → SAMple space reduced
- → Defant in qnertion (mthont replacement)

# Thank You!

**GW Soldiers**